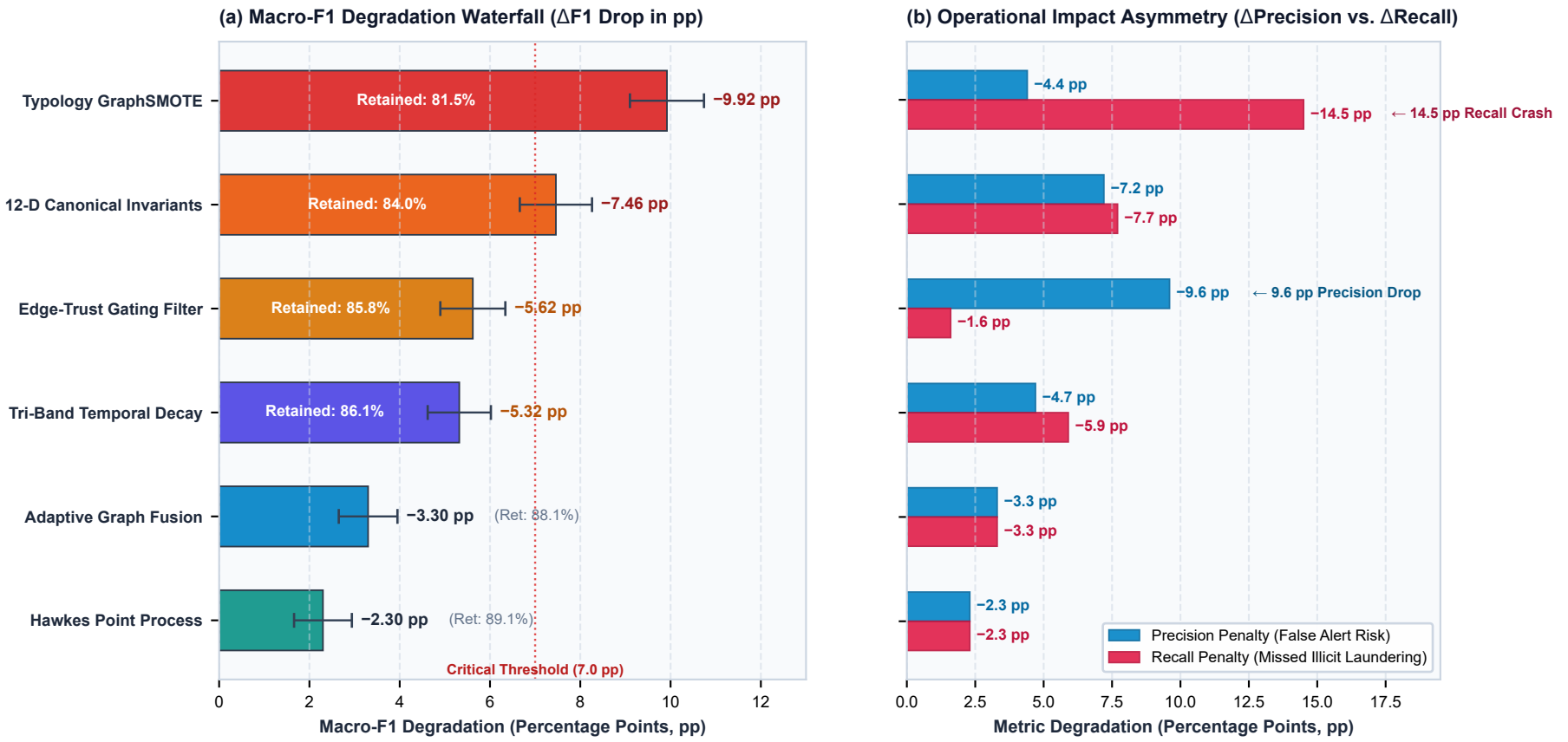
Figure 7.9: Leave-One-Out Component Removal Waterfall & Asymmetry

Marginal degradation on Elliptic-v1 benchmark (N=5 Seeds, Table 7.4 Aligned)

Full C-STGB Benchmark: 91.42 ± 0.65% Macro-F1



Pillar 1: Minority Oversampling

Typology GraphSMOTE

ΔF1: −9.92 pp | ΔRec: −14.5 pp

Synthesizes minority latent motifs; prevents collapse on 0.05% skew.

Pillar 2: Geometric Flow Physics

12-D Invariants z_inv

ΔF1: −7.46 pp | Symmetric ~7.5 pp

Flow balance & arrival kinetics anchor cross-rail transferability.

Pillar 3: Camouflage Denoising

Edge Trust Gating g_ij

ΔF1: −5.62 pp | ΔPrec: −9.6 pp

Prunes evasive smurfing chaff; eliminates false alert fatigue.

Pillar 4: Temporal & View Dynamics

Tri-Band + Hawkes + Fusion

Combined: −10.92 pp | $p < 0.001$

Resolves burst & dormancy scales; calibrates multi-modal evidence.